



Fluid particle diffusion through high-hematocrit blood flow within a capillary tube

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ABSTRACT

Fluid particle diffusion through blood flow within a capillary tube is an important phenomenon to understand, especially for studies in mass transport in the microcirculation as well as in solving technical issues involved in mixing in biomedical microdevices. In this paper, the spreading of tracer particles through up to 20% hematocrit blood, flowing in a capillary tube, was studied using a confocal micro-PTV system. We tracked hundreds of particles in high-hematocrit blood and measured the radial dispersion coefficient. Results yielded significant enhancement of the particle diffusion, due to a micron-scale flow-field generated by red blood cell motions. By increasing the flow rate, the particle dispersion increased almost linearly under constant hematocrit levels. The particle dispersion also showed near linear dependency on hematocrit up to 20%. A scaling analysis of the results, on the assumption that the tracer trajectories were unbiased random walks, was shown to capture the main features of the results. The dispersion of tracer particles was about 0.7 times that of RBCs. These findings provide good insight into transport phenomena in the microcirculation and in biomedical microdevices.

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1. Introduction

Blood is a concentrated suspension of various types of blood cells in plasma, where about 99% of the volume fraction of these different blood cells consists of red blood cells (RBCs). When blood flows in vessels, a micron-scale flow field is generated by RBC motions, which is known to enhance substance diffusion to a level greater than simple Brownian diffusion (Ahuja et al., 1978). Enhancement of mass transport in blood flow is important, because various substances, such as oxygen, nutrients, waste products, and drugs, are transported in the microvascular circulation. The diffusion of platelets is also important, in terms of platelet contact and then adhesion to injured parts of a vascular wall during the first stage of thrombosis formation (Turitto et al., 1972; Wootton and Ku, 1999; Miyazaki and Yamaguchi, 2003). Also, in biomedical microdevices, substances are transported through blood flow within microchannels (Chang et al., 2000; Young and Simmons, 2009). Thus, understanding transport phenomena in microchannels is important not only with regard

to physiological and pathological phenomena, but also for designing biomedical microdevices.

To investigate the role of RBCs in the microcirculation transport phenomena, many researchers have studied the motion and dispersion of RBCs in dilute suspensions flowing through glass capillaries (Goldsmith, 1971a,b; Fischer et al., 1978; Suzuki et al., 1996; Lominadze and Mchedlishvili, 1999; Mchedlishvili and Maeda, 2001; Pries and Secomb, 2003). Because the hematocrit (Hct) in microvessels is around 10–20% (Fung, 1997), it is essential to study blood flow behavior under similar Hct conditions. The main obstacles in tracking RBCs in high-Hct blood are light absorption by hemoglobin and light scattering by RBCs. To overcome this problem, Goldsmith and Turitto (1986) prepared ghost cells by rupturing the RBC membrane and taking out the hemoglobin. Then, they put a small number of RBCs in the suspension of transparent ghost cells and investigated the dispersion of these RBCs. Recently, the authors investigated dispersion of labeled RBCs in concentrated suspensions of healthy RBCs instead of ghost cells (Lima et al., 2008, 2009). Our confocal micro-PTV system enabled the detection of the motions of labeled RBCs even in concentrated suspensions of blood.

While the dispersion of RBCs in blood flow has been investigated widely, there are few reports on the dispersion of fluid particles in blood flow. Goldsmith and coworkers (Goldsmith and Karino, 1977;

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Goldsmith and Marlow, 1979; Goldsmith and Turitto, 1986) investigated tracer particle dispersion in blood flow, but they also used ghost cells, not RBCs, to visualize particle motions in the concentrated suspension. Recently, Gidaspow and Songprawat (2009) measured the diffusion coefficient of nanoparticles also in a suspension of ghost cells. Although these studies are suggestive and interesting, the use of ghost cells instead of RBCs is an important limitation, because the deformation of RBCs changes markedly by varying the viscosity ratio between the intracellular and extracellular fluids of RBCs. In the case of healthy RBCs in plasma, the viscosity of the intracellular fluid, containing hemoglobin, is about 5–8 times larger than that of plasma. In the case of ghost cells, on the other hand, the intracellular and extracellular fluids are identical and the viscosity ratio is unity. Thus, the dispersion of fluid particles in a concentrated suspension of healthy RBCs needs to be clarified.

In this study, the spreading of tracer particles in concentrated suspensions of healthy RBCs flowing in a capillary tube was investigated. By using a confocal micro-PTV system, we could track particle motions in blood with Hct concentrations of up to 20%. We discuss the effect of Hct, radial position of particles, and flow rate, on tracer particle dispersion. We also demonstrate that a scaling analysis could capture the main features of the results.

2. Materials and methods

2.1. Experimental setup

The confocal micro-PTV system used in this study was the same as that used by Lima et al. (2009). For details, please refer our former paper.

2.2. Working fluids

Primarily four kinds of working fluid were used in this study: dextran40 (DX40; Otsoku Pharmaceutical Co., Ltd., Japan) alone and DX40 containing a human RBC Hct of 10%, 15%, and 20%. All samples were seeded with 0.1% (v/v) 1 μm diameter fluorescent tracer particles (F13082, Molecular Probes, Eugene, OR, USA). The density and viscosity of DX40 at 37 °C are $1.046 \times 10^3 \text{ kg/m}^3$ and $3.8 \times 10^{-3} \text{ Pa s}$, respectively. For the comparison of dispersion coefficients between tracer particles and RBCs (section 4.2), we used another fluid: DX40 with 10% Hct, in which 10% of the RBCs were fluorescently labeled with a lipophilic carbocyanine derivative dye, chloromethylbenzamide (CM-Dil, C-7000; Molecular Probes, Eugene, OR, USA) using the same protocol as Ishikawa et al. (in press). To avoid aggregation and adhesion of the particles, the particles and the inner surface of the tube were coated with bovine serum albumin (BSA, 4%; Stock Solution 130-091-376, Miltenyi Biotec., Germany) using the procedure reported by Hou et al. (2009). All procedures in this experiment were carried out in compliance with the Ethics Committee on Clinical Investigation at Tohoku University.

2.3. Experimental procedures

Throughout the experiments, the temperature around the capillary was kept at 37 °C. Hct values were measured by using a hematocrit centrifuge (Kubota 3220, Kubota Corp., Osaka, Japan) just before the experiment. Flow rate was controlled by a syringe pump set to release the working fluid in the range of 0.01–0.07 $\mu\text{L/min}$, which corresponds to a Reynolds number of about 0.001–0.008. Images were taken in the center plane of the capillary at a rate of 100 frames/s. The recorded images were evaluated with a manual tracking plug-in (MTrackj) (Meijering et al., 2006) for the ImageJ software (NIH, Bethesda, MD, USA). Using this software, the position of a bright centroid of a selected tracer particle or a labeled RBC was obtained through successive images.

2.4. Radial dispersion coefficient

The radial dispersion coefficient of particles or RBCs was calculated by (Goldsmith and Marlow, 1979; Goldsmith and Turitto, 1986):

$$D_{rr} = \frac{1}{N} \sum_{i=1}^N \frac{\langle (R_{i,r}(t) - R_{i,r}(0))^2 \rangle}{2t} \quad (1)$$

where $\langle (R_{i,r}(t) - R_{i,r}(0))^2 \rangle$ is the mean square radial displacement of the particles (or RBCs) in the time duration t and N is the number of tracked particles (or RBCs).

If D_{rr} converges to a certain value when t is sufficiently long, the D_{rr} is identical to the diffusion coefficient. In each experimental case, trajectories of more than 100 particles were used for the calculation.

3. Results

3.1. Spreading of tracer particles in 10% Hct blood

Fig. 1 shows the flow of 10% Hct blood with 0.1% tracer particles in the 50 μm diameter glass capillary tube with a flow rate of 0.033 $\mu\text{L/min}$. Fig. 1(a) is the superposition of the confocal image and the normal halogen illumination image, where a cell-free layer was found near the wall. In the pure confocal image (Fig. 1(b)), the tracer particles at the center plane were clearly observed as bright dots, although RBCs were hardly seen. By tracking individual tracer particles, we could measure the trajectories, as shown in Fig. 1(c). Because the flow field was three-dimensional, due to RBC interactions, some particles faded away by moving off the focal plane during the track. In such a case, we simply removed the data of those trajectories, and did not analyze their dispersion.

Dispersion of tracer particles could be calculated from the trajectories, such as those shown in Fig. 1(c), using Eq. (1). The results of the radial dispersion of tracer particles in 10% Hct blood are shown in Fig. 2, where the results of the radial dispersion of tracer particles in DX40 were also drawn for comparison. The dispersion coefficients of both fluids reached almost constant values at about $t = 0.1 \text{ s}$ in this flow rate condition. Thus, the motion of the tracer particles became chaotic after a sufficiently long time period. The dispersion coefficient in DX40 was about $4 \times 10^{-9} \text{ cm}^2/\text{s}$, whereas in 10% Hct blood, it was about $8.4 \times 10^{-9} \text{ cm}^2/\text{s}$. Thus,

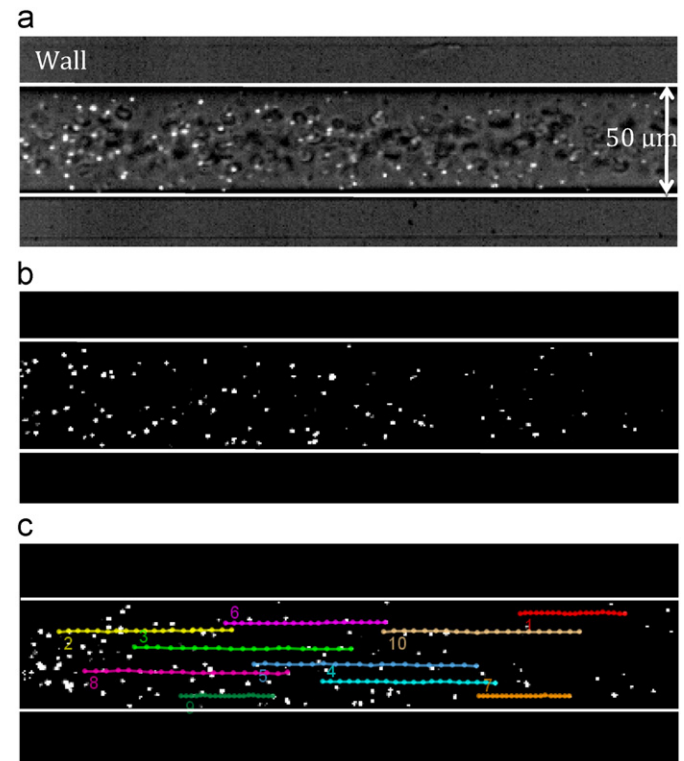


Fig. 1. Recorded images of a 10% Hct sample with 0.1% fluorescent particles in the center plane of a 50 μm glass tube: (a) with both halogen and laser light images superposed, the observed dark rings and bright spots are RBCs and tracer particles, respectively; (b) tracer particles in the confocal image, and (c) with trajectories of tracer particles. The white lines above and below the tracer particles indicate the wall of the capillary tube. The capillary tube has a diameter of 50 μm .

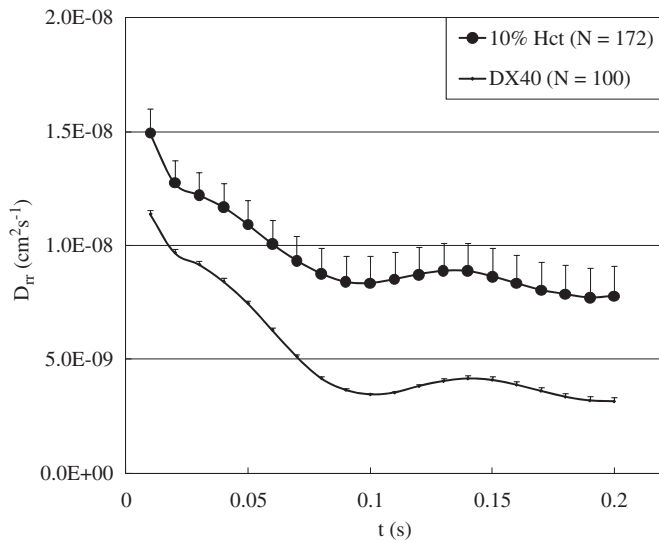


Fig. 2. Comparison of the dispersion coefficient over time between the tracer particle in DX40 and that in 10% Hct blood (flow rate $\sim 0.033 \mu\text{L}/\text{min}$). The error bars indicate the standard deviation according to a t -test analysis at a 95% confidence interval. N is the number of tracked particles.

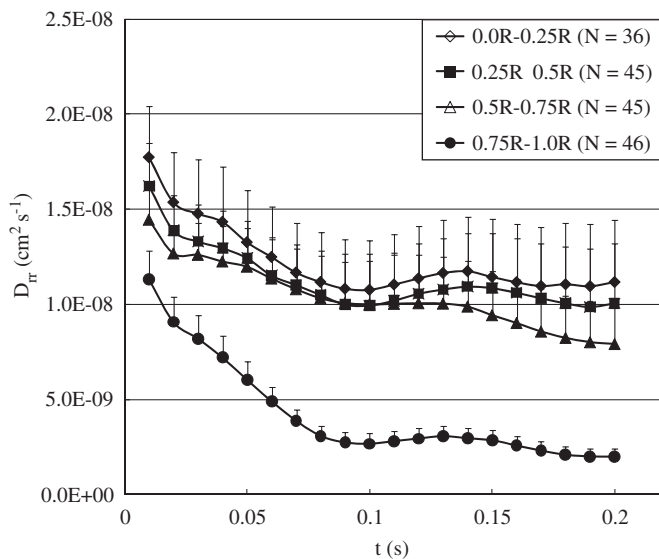


Fig. 3. Tracer particle D_T over time in 10% Hct blood in four different radial positions: 0–0.25R, 0.25–0.5R, 0.5–0.75R, and 0.75–1R, where R is the radius of the tube (flow rate $\sim 0.033 \mu\text{L}/\text{min}$). The error bars indicate the standard deviation according to a t -test analysis at a 95% confidence interval. N is the number of tracked particles.

the presence of RBCs clearly increased the dispersion of the tracer particles in the blood flow. The effects of Brownian diffusion of tracer particles in the system may be evaluated using the Stokes–Einstein equation. By assuming that a rigid sphere with $1 \mu\text{m}$ diameter is suspended in an infinite fluid of DX40 at 37°C the Brownian diffusion could be estimated as $1.2 \times 10^{-9} \text{ cm}^2/\text{s}$, which was slightly smaller than the value we measured in this study.

Moreover, we investigated the radial dependency of the dispersion coefficient. The initial radial position of the particle trajectories was divided into four radial sections with equal length (i.e., 0–0.25R, 0.25–0.5R, 0.5–0.75R, and 0.75–1R, where R is the radius of the tube). Fig. 3 shows the dispersion coefficient of the four different radial sections in 10% Hct blood. It was found that the dispersion coefficient was significantly lower in the 0.75–1R section, compared with the other sections. This was

because there were not many RBCs near the wall due to the formation of cell-free layer (Fig. 1(a)), and the particle movements were restricted by the wall boundary.

3.2. Effect of Hct on the dispersion coefficient

We changed Hct from 10% to 15% and then to 20% to see the effect of Hct on the dispersion of tracer particles. Fig. 4 shows the flow of 20% Hct blood with 0.1% tracer particles with a flow rate of $0.033 \mu\text{L}/\text{min}$. Fig. 4(a) is the superposition of the confocal image and the normal halogen illumination image, in which we saw a higher density of RBCs in the tube and in the thinner cell-free layer than in Fig. 1(a). In the pure confocal image (Fig. 4(b)), the tracer particles at the center plane were again clearly observed as bright dots even at 20% Hct. By tracking individual tracer particles, we could measure the trajectories, as shown in Fig. 4(c). The use of conventional micro-PTV techniques is insufficient to observe tracer particles in such a highly concentrated RBC suspension, due to the increased noise. However, our confocal micro-PTV system enabled us to track particles with high resolution in space and time.

Fig. 5 shows the radial dispersion coefficient of tracer particles with three different Hct (10% Hct, 15% Hct, 20% Hct). These results clearly demonstrate that the dispersion coefficient increased considerably with Hct. Although the error bars in the figure show a large standard deviation, this does not indicate that the results were inaccurate. Since the particle dispersion is strongly dependent on the particle radial position (Fig. 3), the large standard deviation of the dispersion coefficient remains even with large number of tracking particles. We took a time average of the dispersion coefficient from $t = 0.07 \text{ s}$ to the end of the tracking and

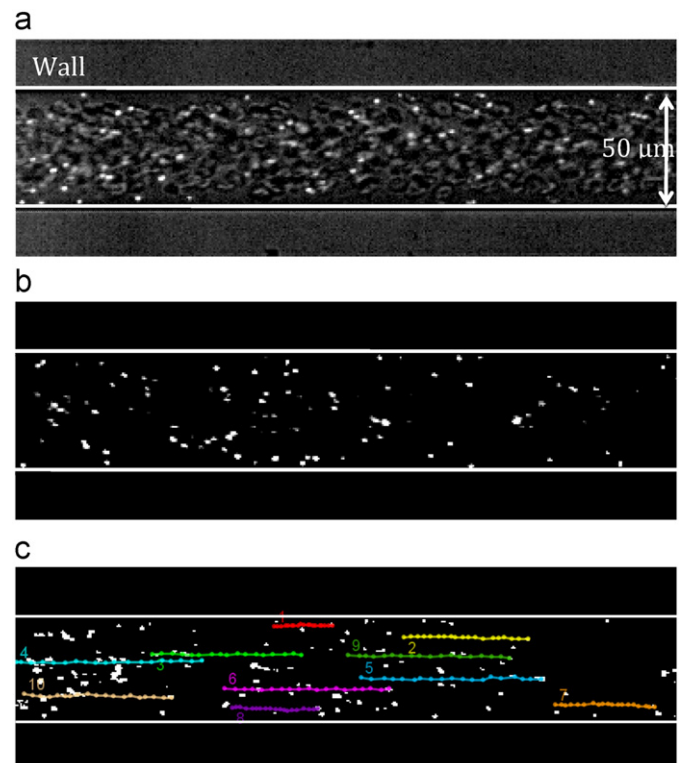


Fig. 4. Recorded images of 20% Hct blood with 0.1% fluorescent particles in the center plane of a $50 \mu\text{m}$ glass tube: (a) with both halogen and laser light images superposed, the observed dark rings and bright spots are RBCs and tracer particles, respectively; (b) tracer particles in the confocal image, and (c) with trajectories of tracer particles. The white lines above and below the tracer particles indicate the walls of the capillary tube. The capillary tube has a diameter of $50 \mu\text{m}$.

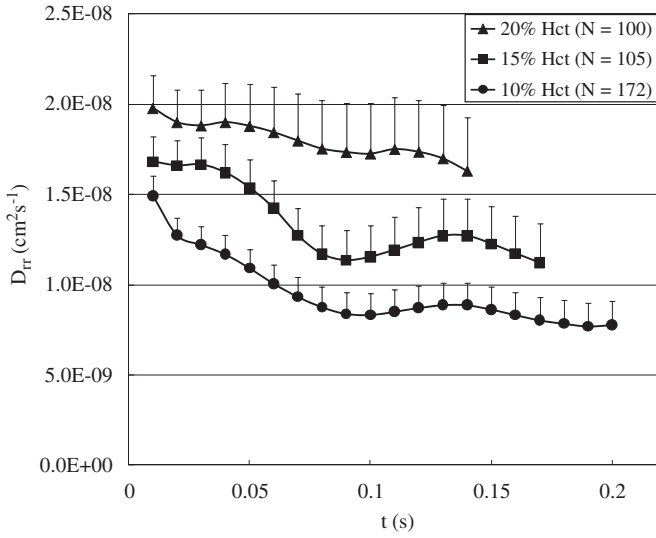


Fig. 5. Effect of Hct on the dispersion coefficient over time (flow rate $\sim 0.033 \mu\text{L}/\text{min}$). The error bars indicate the standard deviation according to a t -test analysis at a 95% confidence interval. N is the number of tracked particles.

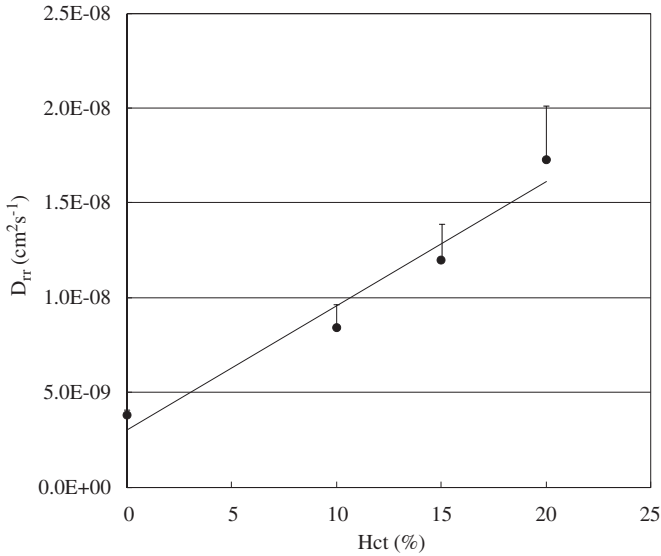


Fig. 6. Effect of Hct on averaged D_{tr} (flow rate $\sim 0.033 \mu\text{L}/\text{min}$). The error bars indicate the standard deviation according to a t -test analysis at a 95% confidence interval.

used this to describe the correlation between Hct and D_{tr} more efficiently, as shown in Fig. 6. We saw an almost linear dependence of averaged D_{tr} on Hct. We further describe this tendency when we used the scaling argument (Section 4.1).

3.3. Effect of flow rate on the dispersion coefficient

To clarify the effect of the flow rate on particle dispersion, the flow rate was varied from about 0.01 to 0.07 $\mu\text{L}/\text{min}$. The averaged dispersion coefficients of 10% Hct blood with various flow rates are shown in Fig. 7. It was found that D_{tr} increased almost linearly with the flow rate. We also describe this tendency further by the scaling argument (Section 4.1). We should note that the dispersion coefficient of tracer particles in DX40 (i.e., 0% Hct) also showed a slight increase with the flow rate. This was caused by the Taylor dispersion effects due to the coupling between Brownian motion and Poiseuille-like flow velocity profile leading

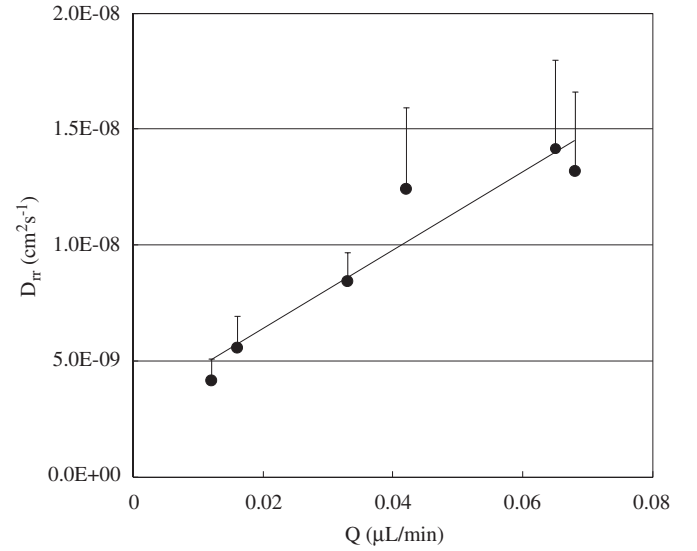


Fig. 7. Effect of flow rate on D_{tr} in 10% Hct blood. The error bars indicate the standard deviation according to a t -test analysis at a 95% confidence interval.

to convective transport, as reported by Taylor (1953). Thus, in Fig. 7, the dispersion of particles was induced primarily by the flow field generated by the RBC interactions, but also by small contributions from the Brownian motion of particles and Taylor dispersion effects.

4. Discussion

4.1. Scaling

We next performed a scaling analysis in an attempt to understand the basic mechanism of the diffusion of tracer particles in the blood flow. Inside the blood flow, particles interact with surrounding RBCs, which changes their orientations and configurations chaotically. Thus, the motion of tracer particles might be expected to resemble a random walk. If we let the length-scale for a particle to drift in the radial direction by colliding to a single RBC be dr , dr may be approximated as a constant value if the RBC does not deform considerably and change its size during the interaction. The direction of dr changes randomly in each collision, because the configuration of RBCs around the tracer particle changes chaotically with time. The time-scale between each collision, dt , depends on the number of RBCs around the tracer particle as well as the relative velocity between the tracer particle and the RBC. The former is proportional to Hct and the latter is proportional to the shear rate. Thus, the time-scale between each collision may be assumed to be inversely proportional to Hct multiplied by $\dot{\gamma}$, where $\dot{\gamma}$ is the shear rate. Because the dispersion coefficient D_{tr} is proportional to the square of the length-scale and inversely proportional to the time-scale, its overall tendency can be given as

$$D_{tr} \propto \frac{dr^2}{dt} \propto dr^2 \text{Hct} \dot{\gamma} \quad (2)$$

To check this scaling, we define the dimensionless dispersion coefficient D_{tr}^* as

$$D_{tr}^* = \frac{D_{tr} - D_B}{\text{Hct} \dot{\gamma} d_{RBC}^2} \quad (3)$$

where D_B is the diffusion due to Brownian motion and d_{RBC} is the diameter of an RBC. Because Brownian diffusion is not considered

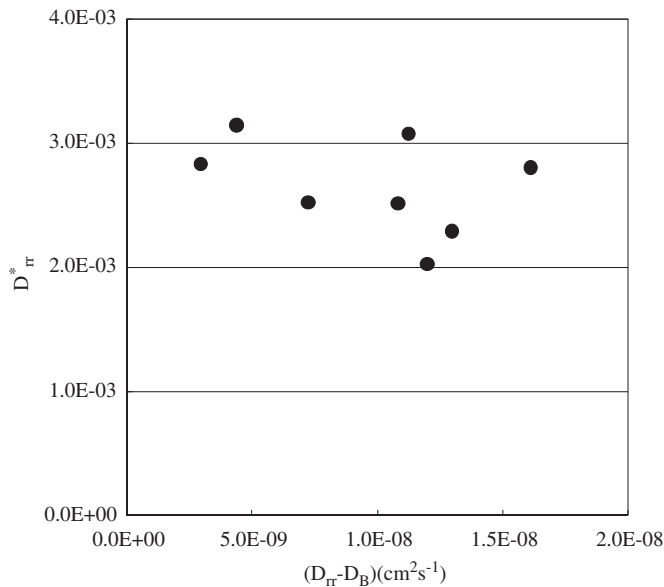


Fig. 8. The results of Figs. 6 and 7 replotted using dimensionless D_{tr}^* .

in the above scaling argument, it is subtracted from D_{tr} . The value of D_B was estimated as $1.2 \times 10^{-9} \text{ cm}^2/\text{s}$ from the Stokes–Einstein equation. d_{RBC} was assumed to be $8 \mu\text{m}$, and $\dot{\gamma}$ was calculated on the wall by assuming the parabolic velocity profile, $\dot{\gamma} = 4U_c/d$, where d is the diameter of the tube and U_c is the center line velocity. In Fig. 8, D_{tr} is plotted against $(D_{tr}-D_B)$ for all experimental data. In the original data of $(D_{tr}-D_B)$, the values are distributed from 2.9×10^{-9} to $1.6 \times 10^{-8} \text{ cm}^2/\text{s}$. The maximum value is about 5.5 times larger than the minimum value. If one rescales the results by Eq. (3), on the other hand, D_{tr}^* values were distributed from 2.0×10^{-3} to 3.1×10^{-3} . Thus, the maximum value was about 1.5 times larger than the minimum value. This result clearly illustrates that the scaling discussed in this section captures the main features of the results.

4.2. Comparison of dispersion coefficients between tracer particles and RBCs

We wanted to examine whether the D_{tr} of tracer particles was larger than that of RBCs or not. The trajectories of labeled RBCs, instead of tracer particles, in 10% Hct blood at the center plane of a $50 \mu\text{m}$ diameter tube were measured. In measuring trajectories of labeled RBCs, there were two major difficulties: (a) small number of labeled RBCs in the blood samples compared to tracer particles, and (b) hard to accurately control the small flow rate for a long time. In order to overcome these difficulties, we performed some experiments under slightly different flow rate conditions with the average of about $0.02 \mu\text{L}/\text{min}$ (in the range of 0.015 – $0.035 \mu\text{L}/\text{min}$), and the results of D_{tr} are averaged over all these cases. This treatment may be verified when the correlation between the flow rate and D_{tr} is linear as shown in Fig. 7. In Fig. 9, D_{tr} of RBCs and tracer particles in 10% Hct blood and D_{tr} of tracer particles in DX40 are shown with the averaged flow rate of about $0.02 \mu\text{L}/\text{min}$. We determined from this figure that the D_{tr} of tracer particles was about 0.7 times that of RBCs in 10% Hct fluid. The statistical analysis of t -test at the 95% confidence interval also confirmed the significant difference between the D_{tr} of particles and RBCs. Thus, fluid particles in suspension do not diffuse as much as RBCs. Goldsmith and Karino (1977) investigated radial displacement of RBCs, white blood cells, $2 \mu\text{m}$ rigid spheres, and $11 \mu\text{m}$ rigid spheres in 40% ghost cell suspensions

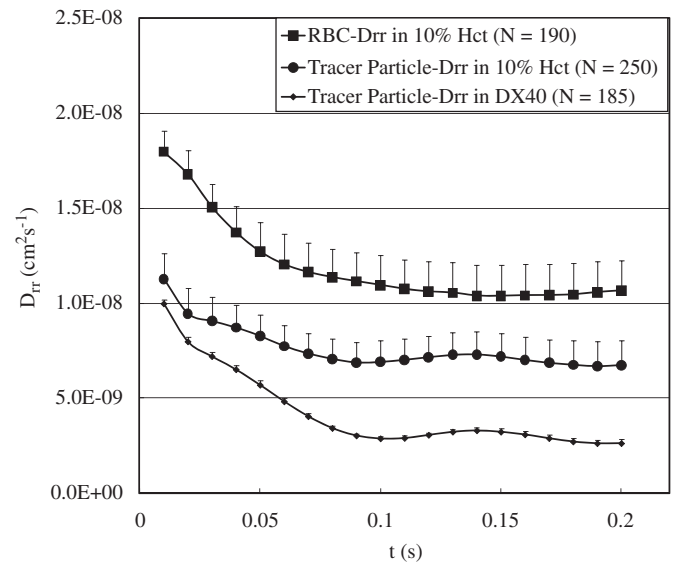


Fig. 9. Comparison of D_{tr} over time between RBCs and tracer particles. RBCs were suspended in 10% Hct blood and tracer particles were suspended in DX40 or 10% Hct blood (averaged flow rate $\sim 0.02 \mu\text{L}/\text{min}$). The error bars indicate the standard deviation according to a t -test analysis at a 95% confidence interval. N is the number of tracked particles (or labeled RBCs).

flowing in a glass capillary tube and although they showed some sample trajectories, they did not discuss the difference in the dispersion coefficients quantitatively.

Although they are not biological suspensions, some researchers have investigated tracer particle diffusion in sheared suspensions of rigid spheres. Breedveld et al. (1998) experimentally investigated the shear-induced diffusion of tracer particles with a diameter of $31.5 \mu\text{m}$ in a suspension of rigid spheres with a diameter of $325 \mu\text{m}$. They found that the diffusion of tracer particles was about 0.7 times of that of rigid spheres. This tendency agrees well with our present results. Others have also investigated fluid particle diffusion in a sheared suspension of rigid spheres computationally (Ishikawa and Yamaguchi, 2008). They found that the diffusion of fluid particles increased linearly with the shear rate, which is also consistent with the present results.

5. Conclusions

In this study, the spreading of tracer particles in blood with up to 20% hematocrit flowing in a capillary tube was studied using a confocal micro-PTV system. We could successfully track hundreds of particles in high-hematocrit blood and measured the radial dispersion coefficient. The results illustrated significant enhancement of the particle diffusion due to a micron-scale flow-field generated by RBC motions. By increasing the flow rate, the particle dispersion increased almost linearly, under a constant hematocrit concentration. The particle dispersion also showed near linear dependency on hematocrit, up to 20%. Scaling of the results on the assumption that the tracer trajectories were unbiased random walks was shown to capture the main features of the results. The dispersion of tracer particles was about 0.7 times that of RBCs. These findings provide helpful insight regarding transport phenomena in both the microcirculation and in biomedical microdevices.

Conflict of interest statement

None declared.

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